

Nonintrusive Parameter Identification of IoT-Embedded Isotropic PMSM Drives

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Abstract—This paper proposes a nonintrusive parameter identification procedure suitable for Internet-of-Things (IoT)-embedded isotropic permanent magnet synchronous machines (PMSM). The method is designed for scenarios where only measurements collected without additional sensors, dedicated tests or signal injection from in-service off-the-shelves motor drives are available. After the automatic detection of the steady state operating conditions defined by the triplet current-speed-temperature, the rotor flux linkage, the stator resistance, the inductance, and the inverter distorted voltage term are estimated using two operating points. Particular emphasis is placed in defining the criteria of selecting these two optimal operating conditions to minimize the estimation errors. The latter are due to the inevitable difference between the parameters in different operating points. As a vessel to investigate the effectiveness of the proposed parameter identification, experimental and simulation tests carried out on a high-speed PMSM drive have been used for validation purpose. The proposed method is also compared with existing methods from the literature to demonstrate its superiority in the considered scenario.

Index Terms—Actuation delay compensation, inverter nonlinearity, parameter identification, permanent magnet, rank deficiency, synchronous machines.

I. INTRODUCTION

Recently, the integration of electrical drives with edge and cloud computing systems is becoming more and more popular for condition monitoring, fault diagnosis and predictive maintenance [1], [2]. In this context, data-driven and machine learning-based algorithms are the most adopted, while model-based (MB) approaches have not received much attention [3]. However, MB strategies may offer additional advantages if paired with data-driven algorithms. A consolidated strategy is to execute an analytical digital twin in parallel to the real motor with the same measured inputs while comparing their outputs. The residual signals can be exploited to detect faults in the system using a residual evaluator.

To implement mathematical models of motors, the parameter identification is essential. Moreover, the knowledge of the parameters allows itself for a straightforward fault diagnosis if the model parameters have a one-to-one mapping with

the physical coefficients [4]. For instance, the estimation of rotor magnet flux of permanent magnet synchronous machines (PMSMs) can be adopted to detect magnet faults [5], while the interturn short circuit can abruptly change both dq -axis inductance and winding resistance [6].

In this study, the problem of the parameter identification of Internet-of-things (IoT)-embedded electrical motor drives is addressed. A schematic of the considered IoT-embedded electrical motor drive is depicted in Fig. 1. The system consists of a standard off-the-shelf motor drive, an IoT device which collects and transmits the data generated by the drive control unit, and a cloud-computing application where the data are stored and processed in order to accomplish parameter identification. The communication between the drive control unit and the IoT device is enabled by the widespread connectivity properties of modern electric drives. The flexibility and scalability of this architecture allows to easily implement customized parameter identification algorithms for multiple and different in-service motor drives. In the considered setup, the drive control unit shares the data with the IoT device through an unidirectional communication. Therefore, the drive control algorithm is not affected by the parameter identification while the estimated parameters can be exploited to implement digital twins, condition monitoring, fault detection, or predictive maintenance procedures.

Thus, this work is based on a realistic scenario where a motor drive provides measurements generated during regular operations. In such a scenario, compared to more conventional contexts such as laboratory tests or electric drive design, additional nonintrusivity and flexibility constraints must be considered to design a suitable parameter identification technique:

- 1) to simplify the system and reduce costs, additional measurement devices are not allowed;
- 2) to avoid machine downtimes, ad hoc tests to collect data are not allowed;
- 3) since in-service off-the-shelves commercial motor drives are considered, a customized control logic cannot be implemented to inject perturbation signals;
- 4) the algorithm must be flexible to operate even in case of few available speed/load operating conditions.

In particular, the parameter identification problem is formulated for PMSMs, which are widely used in the industry [7], electric transportation [8], and renewable energy [9]. The identification of the PMSM parameters, such as inductances,

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rotor flux linkage, and stator resistance has been subject of a vast literature in the last decades. The numerous parameters estimation techniques reported in literature are categorized into two main families: offline and online estimation methods.

Offline methods need specific laboratory equipment with the machine usually disconnected from its load or require multiple measurements collected in different motor operating conditions achieved with ad hoc tests [10]. For instance, in [11], an offline procedure is proposed in which the rotor flux linkage is estimated using line-to-line voltage measurements during a test in which an induction motor is powered to drive the PMSM. Instead, to identify the stator resistance and inductance, an additional single-phase voltage source is connected to the PMSM to collect the required measurements. In [12], no load and load tests on several operating points in the entire torque-speed range are performed to identify the motor parameters. In [13], the measurements are collected under multiple predefined speed and current conditions achieved with ad hoc tests.

Instead, online parameter identification methods are usually implemented on the drive control unit during the motor operation relying on the injection of perturbation signals to overcome the rank-deficiency issue. For instance, in [6], a d -axis current is injected to obtain two steady-state equations required to identify the parameters. In [14], two position offsets are intentionally added into the drive system and the machine measurements corresponding to the two offsets are recorded before the estimation. In [15], an additional temporary steady state is designed for the simultaneous estimation of flux, resistance and dq -axis inductances of a salient-pole PMSM. It is clear that both approaches are inadequate to be adopted in the scenario considered in this work.

A first work that addresses the parameter identification of a PMSM using cloud computing resources was presented in [16]. In this article, an edge/cloud computing architecture based on Amazon Web Services (AWS) is arranged to collect motor measurements generated without signal injection or ad hoc tests and to perform a parameter identification based on three Adaline Neural Networks. However, this work lacks of validation through experimental data collected from a real motor. Moreover, this work is based on simplifying assumptions, such as the availability of the measured voltages and the absence of parameter variations.

In this paper, a novel parameter identification method for isotropic PMSM drives controlled with zero d -axis current is proposed. Note that this is the most widespread typology among PMSM drives and can be also considered the most investigated for the parameter identification [6], [11], [14], [17]–[22]. In particular, to satisfy the nonintrusivity constraints imposed by the considered scenario, the motor parameters are estimated using only measurements commonly available in commercial PMSM drives (such as rotor position, currents, voltage references, and motor temperature) collected during the regular operation of the motor without injection of perturbation signals or ad hoc tests. Moreover, to make the method more flexible and to extend its applicability in contexts where few load/speed operating conditions are available, the identification algorithm is designed to operate only with two

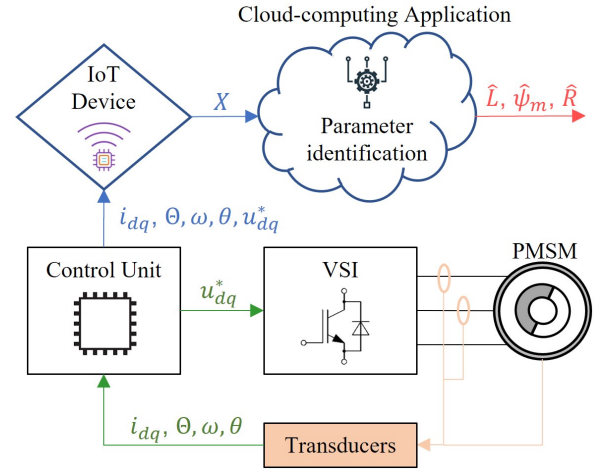


Fig. 1: IoT-embedded drive schematic.

steady states.

Since the motor parameters may vary among the PMSM steady states due to temperature and frequency changes, a theoretical analysis is presented to determine the estimation errors caused by the parameter variations. Therefore, according to this analysis, when only two operating conditions are available, the proposed identification method evaluates the feasibility of the estimations. Instead, when more than two operating conditions are available, an optimization algorithm selects the two operating conditions used to identify the parameters which minimize the estimation errors.

Note that, since the proposed method relies on the voltage references instead of the measured voltages, the estimation accuracy can be further endangered [22]. The mismatch between these two quantities is caused by actuation delays and inverter nonlinearity, i.e., voltage drops on power devices and dead-time effect [23]. The inverter distortion does not affect the estimation of the q -axis inductance when the d -axis current is controlled to zero [23], while the actuation delay affects all the parameters. In particular, the actuation delay causes a voltage distortion which increases when the ratio between electrical rotor speed and control sampling frequency increases. As a consequence, identification procedures relying on voltage references are usually carried out at low speeds where this effect is negligible [6], [20]. In this paper, both inverter nonlinearity and actuation delay are considered and compensated to increase the accuracy also in medium and high frequency applications.

The rest of this paper is organized as follows. Section II summarizes the PMSM model with actuation delays and inverter non-linearity, while the proposed identification algorithm is detailed in Section III. The validation methodology is discussed in section IV while the simulation results and the experimental validation are reported and commented in Section V and VI, respectively.

II. FORMULATION OF THE PARAMETER IDENTIFICATION PROBLEM

A stationary discrete-time mathematical model largely adopted for the design of online identification algorithms of

isotropic PMSM with d -axis current, i_d , equal to zero, is based on the following equations [6], [22], [24]:

$$u_d^*(k) = -L\omega(k)i_q(k) - D_d(k)V_{dead}, \quad (1a)$$

$$u_q^*(k) = Ri_q(k) + \psi_m\omega(k) - D_q(k)V_{dead}, \quad (1b)$$

where k denotes the k -th sample, u_{dq}^* are the voltage references, i_q is the q -axis current, ω is the electrical rotor speed, V_{dead} is the distorted voltage, D_d and D_q are the distorted coefficients as in [5], and L , R , and ψ_m are the stator inductance, the stator resistance, and the rotor flux linkage, respectively. The sampling time of (1) is T_s . This model is based on the common assumptions that the dq -axis inductances are equal and iron losses are negligible in isotropic PMSMs [21], [23].

Since temperature measurements are usually available in commercial PMSM drives [25], the model can be improved by introducing the temperature correction of R :

$$u_d^*(k) = -L\omega(k)i_q(k) - D_d(k)V_{dead}, \quad (2a)$$

$$u_q^*(k) = (1 + \alpha_0\Delta\Theta(k))R'_{AC}i_q(k) + \psi_m\omega(k) - D_q(k)V_{dead}, \quad (2b)$$

where $\Delta\Theta = \Theta - 20^\circ\text{C}$, with Θ motor temperature, α_0 is the copper temperature coefficient and R'_{AC} is the AC stator resistance factor. Note that this PMSM model has four unknown parameters to be identified, i.e., L , R'_{AC} , ψ_m , and V_{dead} , while its rank is at most two. Therefore, in principle, only two PMSM steady states are required to estimate the parameters.

To increase the estimation accuracy, more complex models are considered in offline estimation methods. For instance, in [26], a model which includes also iron losses and self- and cross-saturation is considered. The study presented in [27], shows that the AC stator resistance is actually function of the temperature and frequency by means of additional coefficients and parameters to be identified. Obviously, the increased complexity of the model results in more parameters to be identified and in additional PMSM operating conditions required to solve the parameter identification problem. However, in the considered scenario, the PMSM operating conditions cannot be arbitrarily obtained with ad hoc tests since only measurements collected during the regular operation of the motor can be exploited. Therefore, in this study, the model (2) is chosen as it is simple and flexible to operate even in case of PMSMs working in few different conditions.

The accuracy of (2) is further improved by compensating the delay between the actual voltages and the reference voltages, which is equal to $1.5T_s$ in the PMSM digital control system [28]:

$$\tilde{u}_{dq}(k) = \begin{bmatrix} \cos(\Delta\theta(k)) & \sin(\Delta\theta(k)) \\ -\sin(\Delta\theta(k)) & \cos(\Delta\theta(k)) \end{bmatrix} u_{dq}^*(k-1), \quad (3)$$

where $\Delta\theta$ is the rotor electrical angular displacement due to the digital delay. By assuming constant speed during this short period, $\Delta\theta$ is expressed as:

$$\Delta\theta(k) = \frac{3}{2}(\theta(k) - \theta(k-1)), \quad (4)$$

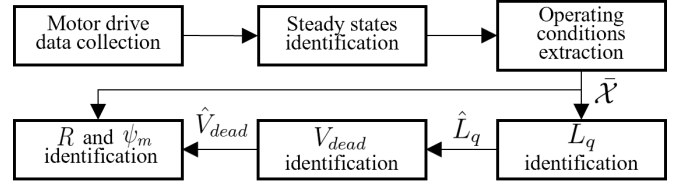


Fig. 2: Main steps of the proposed identification method.

where θ is the rotor electrical position. Note that since $\Delta\theta(k) \approx (3/2)\omega(k)T_s$, the voltage distortion caused by the digital delay is expected to increase with the speed of the motor. By considering (3), the definitive model adopted in this study is:

$$\tilde{u}_d(k) = -L\omega(k)i_q(k) - D_d(k)V_{dead}, \quad (5a)$$

$$\tilde{u}_q(k) = (1 + \alpha_0\Delta\Theta(k))R'_{AC}i_q(k) + \psi_m\omega(k) - D_q(k)V_{dead}. \quad (5b)$$

III. PROPOSED PARAMETER IDENTIFICATION METHOD

Fig. 2 shows the high-level procedure of the proposed parameter identification detailed in the next subsections. The identification of the electromagnetic steady states defined by a given speed and current values is outlined in subsection III-A along with extraction of the operating conditions (OCs) defined by a certain speed, current, and stator temperature. Subsection III-B describes the identification of L and V_{dead} while subsection III-C details the estimations of R'_{AC} and ψ_m . The selection of the best OCs used to estimate R'_{AC} and ψ_m is described in Subsection III-D.

A. Extraction of the operating conditions

The formulated parameter identification problem is based on the stationary model of the PMSM. Therefore, the automated detection of the motor steady states is required. In this study, a steady state is detected when the recorded current and speed show a constant behaviour according to the R-statistic paradigm [16]. Instead, the PMSM transient states due to speed and load variations are automatically excluded from the collected measurements. Let N_s be the number of detected steady states. Then, X_i , with $i = 1, \dots, N_s$, denotes the set containing the data of the i -th steady state, as follows:

$$X_i = \{\omega_i(1), \dots, \omega_i(N_i), i_{q_i}(1), \dots, i_{q_i}(N_i), \tilde{u}_{dq_i}(1), \dots, \tilde{u}_{dq_i}(N_i), D_{d_i}(1), \dots, D_{d_i}(N_i), D_{q_i}(1), \dots, D_{q_i}(N_i), \Theta_i(1), \dots, \Theta_i(N_i)\}, \quad (6)$$

where N_i is the number of samples within the i -th steady state and Θ_i represents the motor temperature. Note that in (6) speed and current are at the steady state while the motor temperature may vary. Since both R and ψ_m depends on the temperature, it is congruous to consider different OCs within each electromagnetic steady state according to thermal transients. Therefore, in each X_i , a set of OCs is extracted as follows.

First, the raw data in each X_i is properly sliced. Let $k_{i,h_s}, k_{i,h_e} \in \{1, \dots, N_i\}$, with $k_{i,h_e} > k_{i,h_s}$, be the start and end sample of the h -th data slice, respectively. k_{i,h_s}

and k_{i,h_e} are such that $|\Theta_i(k_{i,h_s}) - \Theta_i(k_{i,h_e})| \approx \Delta\Theta_r$. That is, each slice is a sub set of the data in X_i corresponding to a temperature range that is approximately $\Delta\Theta_r$. Then, each h -th slice in X_i is represented by the t -tuple $(\bar{\omega}_{i,j}, \bar{i}_{q_{i,j}}, \bar{u}_{dq_{i,j}}, \bar{D}_{d_{i,j}}, \bar{D}_{q_{i,j}}, \bar{\Theta}_{i,j})$ obtained by averaging each variable within the slice.

Now let $\bar{\mathcal{X}} = \{\bar{X}_1, \dots, \bar{X}_{N_{OC}}\}$ be the set of the OCs extracted from all X_i sets. Each OC is represented by $\bar{X}_j = [\bar{\omega}_j, \bar{i}_{q_j}, \bar{u}_{dq_j}, \bar{D}_{d_j}, \bar{D}_{q_j}, \bar{\Theta}_j]$, $j = 1, \dots, N_{OC}$. For each X_i , the t -uples corresponding to the first and last data slices constitute the first two OCs extracted. Then, other OCs are extracted at a temperature step of $\Delta\Theta_s$. That is, if exist two data slices, indexed by $h1$ and $h2$, such that $|\bar{\Theta}_{1,h1} - \bar{\Theta}_{1,h2}| \approx \Delta\Theta_s$, the corresponding t -uples are added as OCs in $\bar{\mathcal{X}}$. Clearly, $\Delta\Theta_s$ is selected greater than $\Delta\Theta_r$.

To sum up, the extraction procedure starts from the raw data of each identified steady state, X_i , to produce the set of OCs, $\bar{\mathcal{X}}$, to be employed in the parameter identification. Two or more OCs in $\bar{\mathcal{X}}$ may share the same steady state values of speed and current but feature different temperatures (with a step of $\Delta\Theta_s$ degrees).

B. L and V_{dead} identification

After the OCs have been extracted, the inductance in each OC, L_j , can be identified using (5a):

$$\hat{L}_j = \frac{-\bar{u}_{d_j}}{\bar{\omega}_j \bar{i}_{q_j}}, \quad (7)$$

where $D_d V_{dead}$ is neglected as it is a periodic function of period $\frac{2\pi}{6\bar{\omega}_j}$ with zero-mean when $i_d = 0$ [20]. This is a simple solution whose main novelty relies on to use of the d -axis delay-compensated voltage, which ensures higher accuracy compared to conventional approaches based on the use of the d -axis voltage reference. It is also worth noticing that, since only one OC is required to estimate L , the variation of this parameter due to saturation effects among the collected OCs can be effectively tracked.

While the VSI nonlinearity does not affect the estimation of L , it affects the estimation of ψ_m and R'_{AC} as the term $D_q V_{dead}$ in (5b) has a non-zero mean value. Therefore, it is necessary to estimate V_{dead} to ensure the accuracy of the estimation of ψ_m and R'_{AC} . The identification of V_{dead_j} can be addressed by solving the following minimization problem for each OC, derived from (5a):

$$\min_{\hat{V}_{dead_j}} \sum_{k=k_{s_{i,h}}}^{k_{e_{i,h}}} (D_{d_i}(k) \hat{V}_{dead_j} + \bar{u}_{d_i}(k) + \hat{L}_j \omega_i(k) i_{q_i}(k))^2, \quad (8)$$

where i and h denotes the i -th steady state and the h -th slice corresponding to the j -th OC in $\bar{\mathcal{X}}$. The effectiveness of the search depends on the availability of samples in a period of $D_d V_{dead}$, which decreases as the motor speed increases since the sample time T_s is fixed. Note that also to identify V_{dead} , only one OC is required.

C. R'_{AC} and ψ_m Identification

To obtain a full-rank identification problem for R'_{AC} and ψ_m , two different OCs, $\alpha, \beta \in \{1, \dots, N_{OC}\}$, are employed. In this way, a set of two equations is derived from (5b):

$$\begin{bmatrix} \bar{u}_{q_\alpha} + \bar{D}_{q_\alpha} \hat{V}_{dead_\alpha} \\ \bar{u}_{q_\beta} + \bar{D}_{q_\beta} \hat{V}_{dead_\beta} \end{bmatrix} = \begin{bmatrix} (1 + \alpha_0 \Delta \bar{\Theta}_\alpha) \bar{i}_{q_\alpha} & \bar{\omega}_\alpha \\ (1 + \alpha_0 \Delta \bar{\Theta}_\beta) \bar{i}_{q_\beta} & \bar{\omega}_\beta \end{bmatrix} \begin{bmatrix} \hat{R}'_{AC} \\ \hat{\psi}_m \end{bmatrix}, \quad (9)$$

where $\hat{R}'_{AC}$ and $\hat{\psi}_m$ are, respectively, the estimated stator resistance and rotor flux linkage. To obtain two linearly independent equations, the OCs α and β must be selected such that:

$$r = \frac{(1 + \alpha_0 \Delta \bar{\Theta}_\alpha) \bar{i}_{q_\alpha} \bar{\omega}_\beta}{(1 + \alpha_0 \Delta \bar{\Theta}_\beta) \bar{i}_{q_\beta} \bar{\omega}_\alpha} \neq 1. \quad (10)$$

If (10) is satisfied, $\hat{R}'_{AC}$ and $\hat{\psi}_m$ are obtained as:

$$\hat{R}'_{AC} = \frac{-\frac{\bar{\omega}_\beta}{\bar{\omega}_\alpha} (\bar{u}_{q_\alpha} + \bar{D}_{q_\alpha} \hat{V}_{dead_\alpha}) + \bar{u}_{q_\beta} + \bar{D}_{q_\beta} \hat{V}_{dead_\beta}}{\bar{i}'_{q_\beta} (1 - r)} \quad (11a)$$

$$\hat{\psi}_m = \frac{\bar{u}_{q_\alpha} + \bar{D}_{q_\alpha} \hat{V}_{dead_\alpha} - (\bar{u}_{q_\beta} + \bar{D}_{q_\beta} \hat{V}_{dead_\beta}) \frac{\bar{i}'_{q_\alpha}}{\bar{i}'_{q_\beta}}}{\bar{\omega}_\alpha (1 - r)}. \quad (11b)$$

where $\bar{i}'_{q_\iota} = (1 + \alpha_0 \Delta \bar{\Theta}_\iota) \bar{i}_{q_\iota}$, with $\iota = \alpha, \beta$.

Considering the parameter dependency on the OCs and the presence of voltage errors due to imperfect digital delay and VSI nonlinearity compensation and measurement errors of speed and currents, (5b) can be rewritten as follows:

$$\bar{u}_{q_\alpha} + \bar{D}_{q_\alpha} \hat{V}_{dead_\alpha} = R'_{AC_\alpha} \bar{i}'_{q_\alpha} + \psi_{m_\alpha} \bar{\omega}_\alpha + \varepsilon_{u_{q_\alpha}}, \quad (12a)$$

$$\bar{u}_{q_\beta} + \bar{D}_{q_\beta} \hat{V}_{dead_\beta} = R'_{AC_\beta} \bar{i}'_{q_\beta} + \psi_{m_\beta} \bar{\omega}_\beta + \varepsilon_{u_{q_\beta}}, \quad (12b)$$

where R'_{AC_α} , R'_{AC_β} , ψ_{m_α} , ψ_{m_β} , $\varepsilon_{u_{q_\alpha}}$, and $\varepsilon_{u_{q_\beta}}$ are the unknown actual values of the stator resistance, rotor flux linkage, and voltage errors in the OCs α and β , respectively. By substituting (12a) and (12b) in (11b), the following equivalent expressions of ψ_m are obtained:

$$\hat{\psi}_m = \psi_{m_\alpha} + \varepsilon_{\psi_{m_\alpha}} + \varepsilon_{\psi_M}, \quad (13)$$

$$\hat{\psi}_m = \psi_{m_\beta} + \varepsilon_{\psi_{m_\beta}} + \varepsilon_{\psi_M},$$

where

$$\begin{aligned} \varepsilon_{\psi_{m_\alpha}} &= \frac{(\psi_{m_\alpha} - \psi_{m_\beta})r}{1 - r} + \frac{(R'_{AC_\alpha} - R'_{AC_\beta}) \frac{\bar{i}'_{q_\alpha}}{\bar{\omega}_\alpha}}{1 - r}, \\ \varepsilon_{\psi_{m_\beta}} &= \frac{(\psi_{m_\alpha} - \psi_{m_\beta})}{1 - r} + \frac{(R'_{AC_\alpha} - R'_{AC_\beta}) \frac{\bar{i}'_{q_\alpha}}{\bar{\omega}_\alpha}}{1 - r}, \\ \varepsilon_{\psi_M} &= \frac{\varepsilon_{u_{q_\alpha}} - \varepsilon_{u_{q_\beta}} \frac{\bar{i}'_{q_\alpha}}{\bar{i}'_{q_\beta}}}{\bar{\omega}_\alpha (1 - r)}. \end{aligned} \quad (14)$$

With the same approach, by substituting (12a) and (12b) in (11a), the following expressions of $\hat{R}'_{AC}$ are obtained:

$$\begin{aligned} \hat{R}'_{AC_\alpha} &= R'_{AC_\alpha} + \varepsilon_{R_\alpha} + \varepsilon_{R_M}, \\ \hat{R}'_{AC_\beta} &= R'_{AC_\beta} + \varepsilon_{R_\beta} + \varepsilon_{R_M}, \end{aligned} \quad (15)$$

where

$$\begin{aligned}\varepsilon_{R_\alpha} &= \frac{R'_{AC_\beta} - R'_{AC_\alpha}}{1-r} + \frac{(\psi_{m_\beta} - \psi_{m_\alpha}) \frac{\bar{\omega}_\beta}{\bar{i}_{q_\beta}}}{1-r}, \\ \varepsilon_{R_\beta} &= \frac{(R'_{AC_\beta} - R'_{AC_\alpha})r}{1-r} + \frac{(\psi_{m_\beta} - \psi_{m_\alpha}) \frac{\bar{\omega}_\beta}{\bar{i}_{q_\beta}}}{1-r}, \\ \varepsilon_{R_M} &= \frac{\varepsilon_{u_{q_\beta}} - \varepsilon_{u_{q_\alpha}} \frac{\bar{\omega}_\beta}{\bar{\omega}_\alpha}}{\bar{i}'_{q_\beta} (1-r)}.\end{aligned}\quad (16)$$

Note that $\hat{R}'_{AC}$ and $\hat{\psi}_m$ are given by the actual values in the two OCs α or β , i.e., ψ_{m_α} , ψ_{m_β} , and R'_{AC_α} , R'_{AC_β} , plus two components of the estimation error. The first component, i.e., $\varepsilon_{\psi_{m_\alpha}}$ and $\varepsilon_{\psi_{m_\beta}}$ in (13), ε_{R_α} and ε_{R_β} in (15), is caused by the mismatch of the two parameters in the two OCs. The second component, i.e., ε_{ψ_M} in (13), and ε_{R_M} in (15), is caused by the voltage errors.

D. Minimization of the estimation errors

The estimation errors depend on the speed, temperature and q -axis current of both OCs α and β , (see (14) and (16)). Therefore, the majorant of the estimation error can be minimized by proper selecting these two OCs. With reference to the estimation of R'_{AC} in the α -th OC, the majorant of the estimation error, i.e., $\tilde{\varepsilon}_{R_{\alpha tot}}$, is derived from (16):

$$\tilde{\varepsilon}_{R_{\alpha tot}} = \tilde{\varepsilon}_{R_\alpha} + \tilde{\varepsilon}_{R_M}, \quad (17)$$

with

$$\begin{aligned}\tilde{\varepsilon}_{R_\alpha} &= \left| \frac{\tilde{R}'_{AC_\beta} - \tilde{R}'_{AC_\alpha}}{1-r} \right| + \left| \frac{(\tilde{\psi}_{m_\beta} - \tilde{\psi}_{m_\alpha}) \frac{\bar{\omega}_\beta}{\bar{i}_{q_\beta}}}{1-r} \right|, \\ \tilde{\varepsilon}_{R_M} &= \frac{|\tilde{\varepsilon}_{u_q}| + |\tilde{\varepsilon}_{u_q} \frac{\bar{\omega}_\beta}{\bar{\omega}_\alpha}|}{|\bar{i}'_{q_\beta} (1-r)|},\end{aligned}\quad (18)$$

where $\tilde{R}'_{AC_\iota}$ and $\tilde{\psi}_{m_\iota}$ are rough estimations of R'_{AC_ι} and ψ_{m_ι} , respectively, with $\iota = \alpha, \beta$. $\tilde{\varepsilon}_{u_q}$ is the supposed voltage error.

The rough estimations are computed using only temperature and speed data, i.e., considering the followings [27], [29], [30]

$$\begin{aligned}\tilde{R}'_{AC_\iota} &= \tilde{R}_{DC0} \frac{\left(1 + \tilde{\beta}_0 \left(\frac{\bar{\omega}_\iota}{2\pi}\right)^2\right) - 1}{(1 + \alpha_0(\bar{\Theta}_\iota - 20))^{\tilde{\gamma}+1}} + \tilde{R}_{DC0}, \\ \tilde{\psi}_{m_\iota} &= \tilde{\psi}_0 (1 + \tilde{\alpha}_{PM0}(\bar{\Theta}_\iota - 20)),\end{aligned}\quad (19)$$

where $\tilde{R}_{DC0}$ and $\tilde{\psi}_0$ are rough estimations of R and ψ_m at zero speed and 20°C, $\tilde{\beta}_0$ is the supposed quadratic frequency coefficient, $\tilde{\gamma}$ is the supposed temperature coefficient of the AC resistance and $\tilde{\alpha}_{PM0}$ is the supposed PMs temperature coefficient. The use of precautionary values for $\tilde{\varepsilon}_{u_q}$, $\tilde{\beta}_0$, $\tilde{\gamma}$, and $\tilde{\alpha}_{PM0}$, ensures that:

$$\tilde{\varepsilon}_{R_{\alpha tot}} > \varepsilon_{R_\alpha} + \varepsilon_{R_M}. \quad (20)$$

In particular, for the choice of $\tilde{\beta}_0$, a conservative assumption is that the AC resistance at rated speed is at most ten times the DC resistance. According to this assumption, it can be assumed that:

$$\left(1 + \beta_0 \left(\frac{\omega_r}{2\pi}\right)^2\right) = 10 \quad (21)$$

where ω_r is the electrical rated speed. Therefore, the following formula is adopted to calculate $\tilde{\beta}_0$:

$$\tilde{\beta}_0 = \frac{9}{\left(\frac{\omega_r}{2\pi}\right)^2}. \quad (22)$$

Furthermore, considering that commonly γ is in the range [0 1], it is precautionary set $\tilde{\gamma} = 0$ [27], [31]. Finally, for the choice of $\tilde{\alpha}_{PM0}$, it should be considered that for PMs, the actual temperature coefficient α_{PM0} is in the range $[-0.2 - 0.02] \text{ } \%/^\circ\text{C}$. Therefore, in this study, it is chosen $\tilde{\alpha}_{PM0} = -0.1 \text{ } \%/^\circ\text{C}$.

To obtain $\tilde{R}_{DC0}$, the OCs α and β are selected as follows:

$$\begin{aligned}[\alpha, \beta] &= \underset{\substack{i \in S_R \\ j=1, \dots, N_{OC} \\ i \neq j}}{\operatorname{argmin}} \frac{|\tilde{\varepsilon}_{u_q}| + \left| \tilde{\varepsilon}_{u_q} \frac{\bar{\omega}_j}{\bar{\omega}_i} \right|}{\left| \bar{i}'_{q_j} \left(1 - \frac{\bar{i}'_{q_i} \bar{\omega}_j}{\bar{i}'_{q_j} \bar{\omega}_i}\right) \right|} \\ \text{s.t.} \quad & \left| \frac{\bar{i}'_{q_i} \bar{\omega}_j}{\bar{i}'_{q_j} \bar{\omega}_i} \right| > r_{lim},\end{aligned}\quad (23)$$

where r_{lim} is a positive tuning parameter that ensures to reduce estimation errors (see (16)) and S_R is a set of OCs determined as follows:

$$\begin{aligned}S_R &= \{i = 1, \dots, N_{OC} : \frac{|\bar{\omega}_i^2 - \bar{\omega}_w^2|}{\bar{\omega}_w^2} < f_{lim}\} \\ [w] &= \underset{t=1, \dots, N_{OC}}{\operatorname{argmin}} |\bar{\omega}_t|\end{aligned}\quad (24)$$

in which f_{lim} is a positive dimensionless tuning coefficient.

Problem (23) is built given that $\tilde{R}_{DC0}$ is computed by correcting the resistance estimation in α , i.e., $\tilde{R}'_{AC_\alpha}$. Thus, it is convenient to find the OC α such that $\tilde{R}'_{AC_\alpha}$ is close to $\tilde{R}_{DC0}$, i.e., $\bar{\Theta}_\alpha$ near 20°C and $\bar{\omega}_\alpha$ near zero. While the temperature correction is more reliable as the actual temperature coefficient of the AC resistance γ varies little around the value $\tilde{\gamma} = 0.5$ [27], this is not true for the frequency correction as the actual frequency coefficient β_0 largely depends on the machine. Therefore, to not add conservativeness to (23), it is only considered that $\bar{\omega}_\alpha$ is minimized (see (24)). To increase the accuracy of $\tilde{R}_{DC0}$, also the estimation errors $\tilde{\varepsilon}_{R_\alpha}$ and $\tilde{\varepsilon}_{R_M}$ have to be minimized. $\tilde{\varepsilon}_{R_\alpha}$ cannot be explicitly minimized as the parameter estimations are not known. However, as in (18), $\tilde{\varepsilon}_{R_\alpha}$ can be made small by selecting two OCs α and β which make $|1-r|$ big (second constraint in (23)). Instead, $\tilde{\varepsilon}_{R_M}$ is minimized by considering it as objective function. Given α and β from (23), $\tilde{R}'_{AC_\alpha}$ is computed according to (11a). Finally, a temperature and frequency correction provides $\tilde{R}_{DC0}$:

$$\tilde{R}_{DC0} = \tilde{R}'_{AC_\alpha} \frac{(1 + \alpha_0(\bar{\Theta}_\alpha - 20))^{\tilde{\gamma}+1}}{\tilde{\beta}_0 \left(\frac{\bar{\omega}_\alpha}{2\pi}\right)^2 + (1 + \alpha_0(\bar{\Theta}_\alpha - 20))^{\tilde{\gamma}+1}}. \quad (25)$$

With similar considerations, to obtain $\tilde{\psi}_0$, the OCs α and β are selected as follows:

$$\begin{aligned}[\alpha, \beta] &= \underset{\substack{i=1, \dots, N_{OC} \\ j \in S_\psi \\ i \neq j}}{\operatorname{argmin}} \frac{|\tilde{\varepsilon}_{u_q}| + \left| \tilde{\varepsilon}_{u_q} \frac{\bar{i}'_{q_i}}{\bar{i}'_{q_j}} \right|}{\left| \bar{\omega}_i \left(1 - \frac{\bar{i}'_{q_i} \bar{\omega}_j}{\bar{i}'_{q_j} \bar{\omega}_i}\right) \right|} \\ \text{s.t.} \quad & \left| \frac{\bar{i}'_{q_i} \bar{\omega}_j}{\bar{i}'_{q_j} \bar{\omega}_i} \right| > r_{lim}.\end{aligned}\quad (26)$$

where S_ψ is a set of OCs determined as follows:

$$S_\psi = \{j = 1, \dots, N_{OC} : |\bar{\Theta}_j - \bar{\Theta}_w| < \Theta_{lim}\} \quad (27)$$

$$[w] = \underset{t=1, \dots, N_{OC}}{\operatorname{argmin}} |\bar{\Theta}_t|$$

where Θ_{lim} is a positive tuning parameter. Given α and β from (26), the estimation of ψ_m in the OC β , i.e., $\hat{\psi}_{m\beta}$, is computed according to (11b). Finally, a temperature correction provides $\hat{\psi}_0$:

$$\hat{\psi}_0 = \frac{\hat{\psi}_{m\beta}}{1 + \tilde{\alpha}_{PM0}(\bar{\Theta}_\beta - 20)}. \quad (28)$$

The majorant of the estimation error of ψ_m in the β -th OC, $\tilde{\epsilon}_{\psi_{\beta tot}}$, can be similarly computed with the procedure outlined for R'_{AC} . Once the rough estimations are obtained, the error majorants can be explicitly computed and minimized. For each OC $\alpha \in \{1, \dots, N_{OC}\}$ in which $\hat{R}'_{AC\alpha}$ has to be determined, called main OC (MOC), an OC β , called auxiliary OC (AOC), is chosen to minimize $\tilde{\epsilon}_{R_{\alpha tot}}$, i.e.,

$$\beta = \underset{\substack{i=1, \dots, N_{OC} \\ i \neq j}}{\operatorname{argmin}} \tilde{\epsilon}_{R_{\alpha tot}} \quad (29)$$

$$s.t. \quad r < \varepsilon_{r1} \text{ or } r > \varepsilon_{r2}, \text{ and } \tilde{\epsilon}_{R_{j tot}} < x_R \tilde{R}_{\alpha}.$$

where $\varepsilon_{r1} < 1$ and $\varepsilon_{r2} > 1$ are two tunable thresholds which avoid choosing r close to 1, while x_R is a tunable value between 0 and 1. Note that an estimation in an OC α can be rejected if does not exist an AOC so that $\tilde{\epsilon}_{R_{\alpha tot}}$ is less than a given fraction of $\hat{R}'_{AC\alpha}$, i.e. $x_R \hat{R}'_{AC\alpha}$. Once the AOC is found, α and β are set and the estimation is obtained using (11a). Optimization (29) can also be applied to estimate ψ_m in each OC $\beta \in \{1, \dots, N_{OC}\}$ by substituting β , $\tilde{\epsilon}_{R_{\alpha tot}}$, and $\hat{R}'_{AC\alpha}$, with α , $\tilde{\epsilon}_{\psi_{\beta tot}}$ and $\hat{\psi}_{m\beta}$, respectively. It is worth noticing that, according to (29), when only two OCs are available, the proposed method evaluates the reliability of the achievable estimation. Instead, when more than two OCs are available, the proposed algorithm selects the best AOC that minimizes the error majorant.

IV. VALIDATION METHODOLOGY

The proposed identification procedure is validated on a challenging case study consisting of a custom high-speed PMSM drive with large parameter variations. The specifications of the motor drive are reported in Table I while the setup is shown in Fig. 3. Note that the reported stator resistance value is obtained using an ohmmeter at the room temperature while the stator inductance and rotor flux linkage are derived with the experimental procedure reported in [32], which matches the finite element analysis presented in [33]. The stator inductance can be considered constant as this particular machine is not affected by magnetic saturation. The high-speed PMSM is connected to a load motor via a gearbox and a torque sensor and a custom three-phase VSI has been used to supply the motor. The field-oriented control ($i_d = 0$) has been implemented on a Xilinx Zynq7020 SoC custom control platform.

The next section reports the results obtained considering different scenarios derived from a simulation dataset obtained using a MATLAB/Simulink model of this motor drive. The

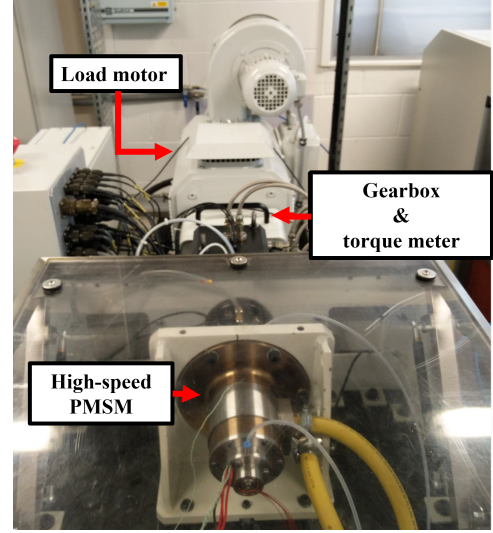


Fig. 3: Experimental setup.

full experimental dataset considered to validate the proposed method is made up of measurements collected during the efficiency characterization of this machine [33]. This dataset, $\mathcal{X}$, consists of 20 different steady states with high temperature, current and speed variations. 55 OCs are extracted from $\mathcal{X}$ by adopting the procedure described in subsection III-A and by setting $\Delta\Theta_r = 1^\circ\text{C}$ and $\Delta\Theta_s = 15^\circ\text{C}$. Fig. 4 reports the main variables during the test at 30 krpm. Fig. 4(a) shows the winding temperature. Two temperature points (77°C and 118°C) corresponding to OCs 52 and 55 are highlighted with the marker. Note that, as shown in Fig. 4, the average value of i_q in these two OCs is the same. Fig. 4(b) reports the measured and reference currents corresponding to OC 52 while Fig. 4(c) shows the voltage references corresponding to OCs 52 and 55. As expected, only the q -axis voltage references are affected by the temperature variation due to changes in ψ_m and R .

The simulation dataset reproduces the same 55 OCs of the experimental dataset allowing the validation of the proposed method in an extended operating range of the motor. In fact, note that the actual values of L , ψ_m , R , V_{dead} , and dq -axis voltages are difficult to measure in all the experimental OCs. In the simulation model, the parameters dependency on the OCs is implemented as follows:

$$R'_{AC} = R_{DC0} \frac{(R'_{AC|20}) - 1}{(1 + \alpha_0(\bar{\Theta}_t - 20))^{\gamma+1}} + R_{DC0}, \quad (30)$$

$$\psi_m = \psi_{m0} (1 + \alpha_{PM0}(\bar{\Theta} - 20)),$$

TABLE I: Motor drive parameter

Parameter	Motor drive	Unit
Rated power	4.20	kW
Rated speed	80	krpm
Rated Current	7	A
DC link voltage	540	V
Number of pole pairs	2	
R_{DC0}	0.67	Ω
ψ_{m0}	26.82	mWb
L	1.25	mH
α_{PM0}	-0.035	$\%/^\circ\text{C}$
Sampling time (T_s)	25	μs

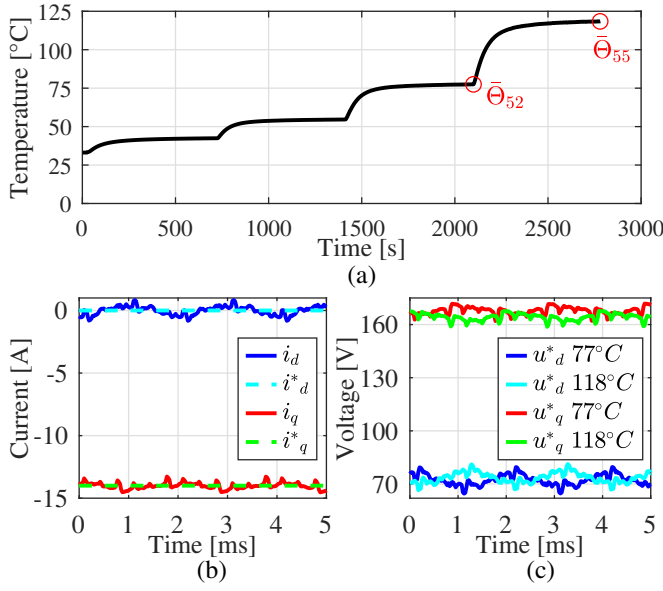


Fig. 4: Measurements during the test at 30krpm of motor drive 1: (a) temperature, (b) dq -axis actual and reference currents at 77°C , (c) dq -axis voltage references at 77°C and 118°C .

where $R'_{AC}|_{20}$ is a 1-D look-up table containing the experimental AC stator resistance measurements at 20°C , R_{DC0} is the experimental measurement value of the DC stator resistance at 20°C and $\gamma = 0.75$. Note that the value of β_0 that fits the experimental measurements $R'_{AC}|_{20}$ is approximately equal to $3.52 \cdot 10^{-7}/\text{Hz}^2$. The inverter parameters ($V_{dead} = -0.35\text{V}$) are considered constant during the simulations as well as the inductance which does not significantly depend on the q -axis current in the investigated case study, according to measurements performed on the real motor. The other parameters required by the proposed procedure are set as follows: $\varepsilon_{u_q} = 0.5\text{V}$, $\beta_0 = 1.27 \cdot 10^{-6}/\text{Hz}^2$, $r_{lim} = 2$, $\Theta_{lim} = 20^\circ\text{C}$, $f_{lim} = 2$, $\varepsilon_{r_1} = 0.9$, $\varepsilon_{r_2} = 1.1$, and $x_R = 0.25$.

V. SIMULATION RESULTS

A. Identification of L and V_{dead}

This subsection presents the results of the identification of L and V_{dead} in all the extracted 55 OCs. Fig. 5 reports with the colour scale in the current-speed-temperature space the actual estimation absolute percentage error (APE) of L defined as $\varepsilon_{L_\alpha} = 100 \left| \frac{L_\alpha - L_\alpha^*}{L_\alpha^*} \right|$. In the same figure, $\varepsilon_{L_\alpha}^*$ represents the APE related to the estimations obtained without compensating the digital delay according to (3). The identification of L using the compensation of the digital delay is highly accurate with a mean APEs (MAPEs) of 0.51% and a maximum error $\varepsilon_{L_{46}} = 2.75\%$. Also the standard deviation (SD) of the estimation error is low (0.68%). Instead, as expected, the results obtained without the compensation of the digital delay are affected by large errors which increase with the motor speed.

To demonstrate the stator inductance tracking ability under saturation effects of the proposed algorithm, an additional analysis is reported. The dependency of L on the q -axis current has been introduced in the simulation model to take into account the saturation effect. The upper subfigure in Fig. 6 shows the q -axis current before and after an abrupt

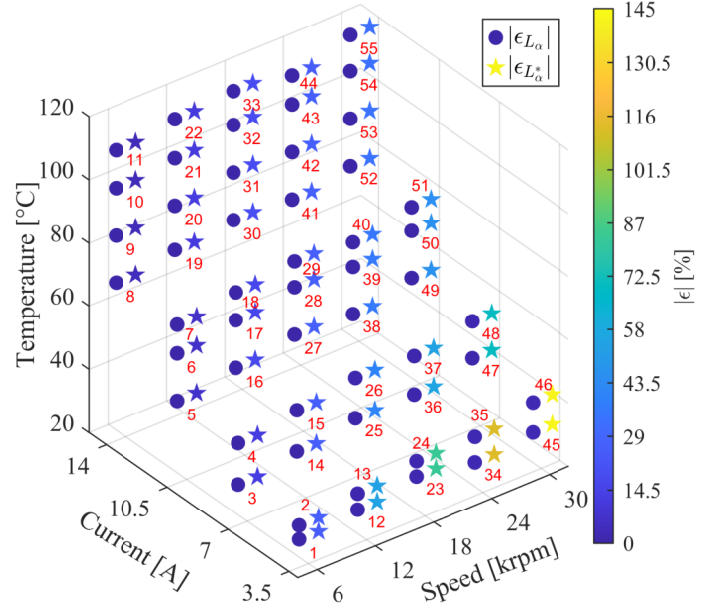


Fig. 5: Estimation error of L .

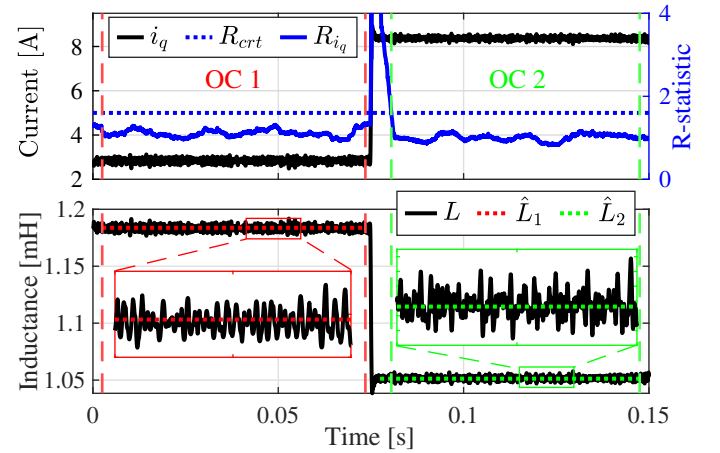


Fig. 6: Estimation of L under saturated conditions.

load torque variation. Thanks to the R-statistic algorithm, the two different steady states of i_q can be easily detected and two different OCs can be extracted. In fact, before the load torque change, the R-statistic index (R_{i_q}) is below the critical threshold (R_{crt}) which means that i_q is at the steady state [16]. The measurements in this time interval constitute the first OC. When i_q arises due to the load torque change, the R-statistic algorithm timely detects the loss of steady state ($R_{i_q} > R_{crt}$). The measurements in this time window can therefore be excluded from the dataset used to estimate the parameters. Finally, R_{i_q} quickly approaches the unity when the steady state is recovered and a new OC is extracted. Note that, to correctly extract the OCs, the R-statistic procedure is also applied on the rotor speed, as mentioned in Subsection III-A. In the lower subfigure, the trend of the actual stator inductance is reported along with its estimations in the two extracted OCs. According to the saturation effect, the actual stator inductance decreases with the increase of the current.

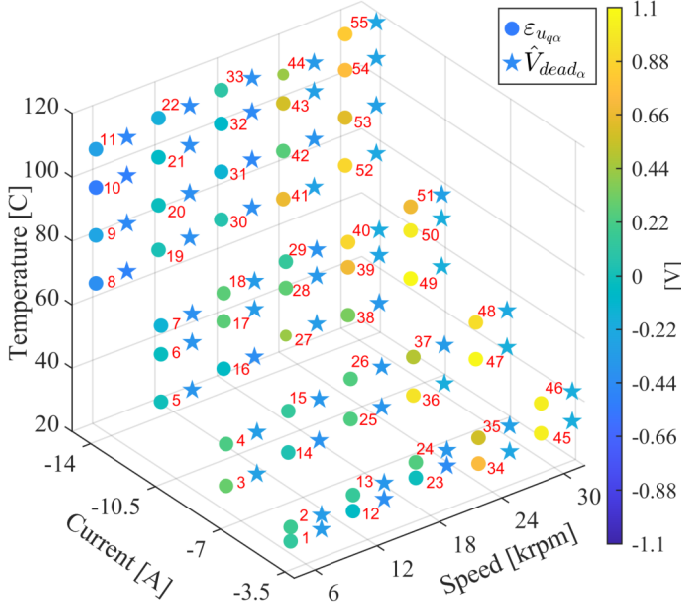


Fig. 7: Voltage error and identified distorted voltage term.

Albeit its variation, the inductance in the two different OCs is tracked very precisely by the proposed algorithm. The reason behind this is that, since from (7) only a single OC is required to estimate L , the accuracy of the proposed algorithm is not affected by inductance variations due to saturation effects. In fact, as i_q is constant within the two OCs, also L is constant. It can be concluded that, in presence of saturation effects, the proposed method can track the inductance variation by estimating the inductance in OCs with different currents.

Fig. 7 reports with the colour scale the q -axis voltage error, $\varepsilon_{u_{q\alpha}}$, and $\hat{V}_{dead\alpha}$, ($\alpha = 1, \dots, 55$), in the current-speed-temperature space. $\varepsilon_{u_{q\alpha}}$ has been computed as:

$$\varepsilon_{u_{q\alpha}} = \bar{u}_{q\alpha} + \bar{D}_{q\alpha} \hat{V}_{dead\alpha} - \bar{u}_{q\alpha}, \quad \alpha = 1, \dots, 55, \quad (31)$$

where $\bar{u}_{q\alpha}$ is the average actual q -axis voltage. The Mean Absolute Error (MAE) of $\varepsilon_{u_{q\alpha}}$ is 0.40V while the maximum error is $\varepsilon_{u_{q49}} = 1.08V$. Note that $\varepsilon_{u_{q\alpha}}$ increases as the speed increases since fewer samples are available in a period of $D_d V_{dead}$. The average value of $\hat{V}_{dead\alpha}$ is $-0.34V$, which is really close to the actual value, and the SD is 2.5%. Therefore, the proposed method provides accurate estimations of the actual q -axis voltage in a wide range of OCs.

B. Identification of R'_{AC} and ψ_m

This subsection present the results of the identification of R'_{AC} and ψ_m considering different scenarios. Fig. 8 shows the results obtained considering the first 5 five steady states, which correspond to the first 13 OCs shown in Fig. 5. In this figure, $\epsilon\% \psi_\alpha$ and $\epsilon\% R_\alpha$ represent the actual APEs, computed similarly to $\epsilon\% L_\alpha$; $\epsilon\% \psi_\alpha = 100 \left| \frac{\psi_\alpha - \psi_M}{\psi_\alpha} \right|$ and $\epsilon\% R_\alpha = 100 \left| \frac{R_\alpha - R_M}{R_\alpha} \right|$ are the theoretical APEs computed respectively using (14) and (16) with the actual values of the parameters and q -axis voltage errors; $\tilde{\epsilon}\% \psi_\alpha = 100 \left| \frac{\tilde{\psi}_\alpha^{tot}}{\psi_\alpha} \right|$ and $\tilde{\epsilon}\% R_\alpha = 100 \left| \frac{\tilde{R}_\alpha^{tot}}{R_\alpha} \right|$ are the minimized absolute percentage values of the error majorants, obtained according to (29). Note that $\epsilon\% \psi_\alpha \approx \tilde{\epsilon}\% \psi_\alpha$ and $\epsilon\% R_\alpha \approx \tilde{\epsilon}\% R_\alpha$ in all the OCs. This outcome confirms the

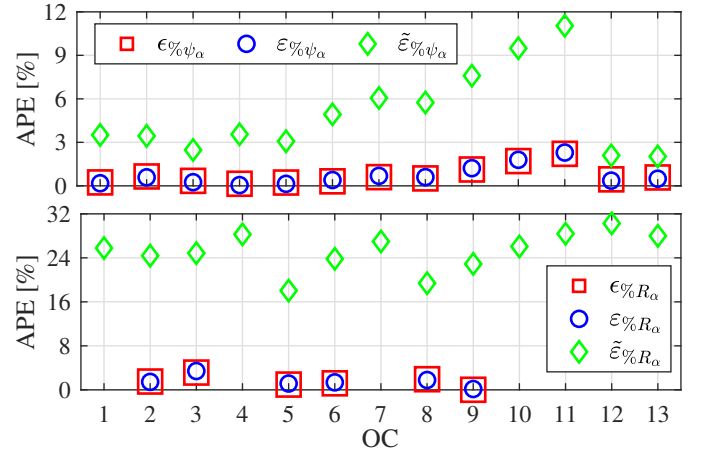


Fig. 8: Results with a set of 5 steady states with $\tilde{\varepsilon}_{u_q} = 0.5V$.

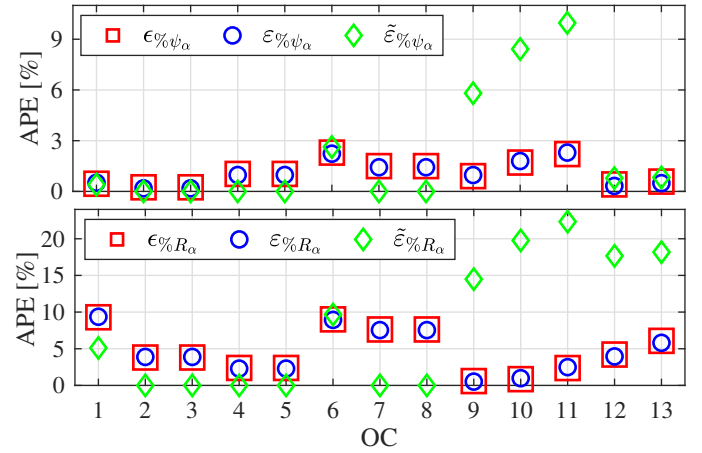


Fig. 9: Results with a set of 5 steady states with $\tilde{\varepsilon}_{u_q} = 0V$.

validity of the error analysis performed in the previous section. In addition, $\tilde{\epsilon}\% \psi_\alpha > \epsilon\% \psi_\alpha$ and $\tilde{\epsilon}\% R_\alpha > \epsilon\% R_\alpha$ in all the OCs, confirming the correct definition of the error majorant. Note that, in some OCs, the actual and theoretical APE of R'_{AC} are not shown as in these cases the estimations have been rejected according to (29). Overall, low estimation errors have been obtained even if the results are asymmetric for the two parameters. In fact, R'_{AC} is the most difficult parameter to be identified due to its high dependency on the OCs.

Fig. 9 shows the results obtained considering the same set of OCs but with $\tilde{\varepsilon}_{u_q} = 0V$, i.e., the voltage error is not considered for the minimization of the error majorant in (29). Note how the estimation accuracy is overall deteriorated for both ψ_m and R'_{AC} and that the error majorant is lower than the actual error in several OCs. It is clear that, even if the q -axis voltage error $\varepsilon_{u_{q\alpha}}$ is small, as shown in Fig. 7, it largely affects the estimation accuracy and must be taken into account for the minimization of the error majorant.

Fig. 10 shows a performance analysis of the identification of ψ_m and R'_{AC} considering several scenarios with different number of available steady states. For each scenario s , N_s different trials produced by randomly choosing k_s steady states among $\mathcal{X}$, with $k_s = 2, \dots, 10$. Note that, since the full dataset contains 20 steady states, the number of steady states

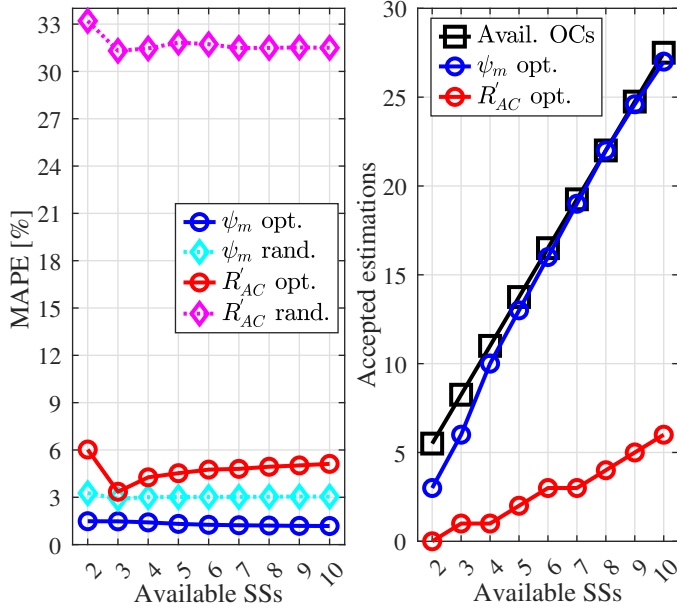


Fig. 10: Identification of ψ_m and R'_{AC} varying the available steady states: (a) MAPE of the estimation results, (b) average number of accepted estimations compared with the average number of available OCs.

combinations without repetition N_{comb_s} when k_s steady states are considered, is given by the following formula:

$$N_{comb_s} = \frac{20!}{k_s!(20 - k_s)!} \quad (32)$$

In this study, to analyze a congruous number of combinations for each value of k_s , it is set $N_s = N_{comb_s}/5$. The analysis considers two performance indexes: the MAPE and the number accepted estimations, both averaged over the N_s trials. Fig. 10(a) shows the MAPE for both ψ_m and R'_{AC} estimations. Despite the challenging case study, the estimation accuracy of ψ_m (blue line) is very high and slightly improves with the increase of the available steady states. This can be explained considering that the availability of more steady states increases the chance to further minimize the majorant of the estimation errors. Also, the estimation error of R'_{AC} (red line) is small even if it is significantly greater than the estimation error of ψ_m . Note that, in this case, while the average estimation accuracy noticeably improves when going from 2 to 3 steady states, subsequently it slightly deteriorates with the increase of the available steady states. This can be explained considering the results reported in Fig. 10(b). This figure shows that, while the estimations of ψ_m are accepted almost in all the available OCs (the blue line is near to the black line), many estimations of R'_{AC} are rejected. Therefore, with reference to the estimation of R'_{AC} , on average, the increase of the available steady states may lead to accept estimations previously rejected which can increase the MAPE. It is worth noticing that, when only 2 steady states are available, on average 4 and 0.25 estimations of ψ_m and R'_{AC} are accepted, respectively. This means that, with 2 steady states, at least an accurate estimation of the rotor flux linkage is guaranteed while the probability to perform reliable estimations of the

stator resistance is around 25%. To better demonstrate how the proposed selection of the AOC improves the accuracy of the parameter estimation, Fig. 10(a) also reports the results obtained by means of a random selection of the AOC (cyan and magenta lines). In particular, for each trial, the AOC is randomly selected among the available OCs that satisfy the condition $r < \varepsilon_{r1}$ or $r > \varepsilon_{r2}$, as in (29). This condition prevent to select a value of r close to 1, which would certainly increase the estimation errors according to (14) and (16). Compared with the estimations obtained with the proposed optimization of the error majorant, the random selection of the AOC leads to a lower estimation accuracy of both parameters. This is particularly clear when looking at the estimation of R'_{AC} . In fact, since this is the parameter most affected by the variation of the OCs, the careful selection of the AOC is crucial.

C. Comparison with the state-of-the-art

In this subsection, the proposed method is compared with other methods from the literature. Since in the context considered in this paper, signal injections and dedicated tests are not possible, two alternative approaches are considered. The first one is based on fixing a parameter to its nominal value while the other parameter is estimated using only one OC [5], [10], [17], [18]. Hereafter, this approach is denoted as fixed parameter (FP) method. When R is fixed to its nominal value R_{DC0} , the estimation of ψ_m in the OC α is provided by:

$$\hat{\psi}_{m\alpha} = \frac{\bar{u}_{q\alpha} + \bar{D}_{q\alpha} \hat{V}_{dead\alpha} - \bar{i}'_{q\alpha} R_{DC0}}{\bar{\omega}_{\alpha}} \quad (33)$$

Note that, as in [5], since the current $\bar{i}'_{q\alpha} = (1 + \alpha_0 \Delta \bar{\Theta}_{\alpha}) \bar{i}_{q\alpha}$ is used to estimate ψ_m , (33) takes into account the variation of R with the temperature. Instead, when ψ_m is fixed to its nominal value ψ_{m0} , R'_{AC} can be identified as follows:

$$\hat{R}'_{AC\alpha} = \frac{\bar{u}_{q\alpha} + \bar{D}_{q\alpha} \hat{V}_{dead\alpha} - \bar{\omega}_{\alpha} \psi_{m0}}{\bar{i}'_{q\alpha}} \quad (34)$$

It is worth highlighting that the accuracy of this method does not depend on the quantity of available steady states as only the data of a single OC are required to estimate the parameters.

The second method is based on a more complex model of the PMSM involving more parameters to explicitly express the dependency of R'_{AC} and ψ_m on the OCs. These parameters are identified by means of the least squares (LS) method exploiting simultaneously all the available OCs, as in [13] and [26]. The following model derived from (9) and (19) is considered for the generic OC α :

$$\bar{u}'_{q\alpha} = \bar{i}'_{q\alpha} R_{DC0} + g_{\alpha}(\gamma', \beta'_0) + \bar{\omega}_{\alpha} \psi_{m0} + \bar{\omega}_{\alpha} (\bar{\Theta}_{\alpha} - 20) \alpha'_{PM0}, \quad (35)$$

with $\bar{u}'_{q\alpha} = \bar{u}_{q\alpha} + \bar{D}_{q\alpha} \hat{V}_{dead\alpha}$, $\bar{\omega}'_{\alpha} = \bar{\omega}_{\alpha}/2\pi$, $\beta'_0 = R_{DC0}\beta_0$, $\gamma' = \gamma + 1$, $\alpha'_{PM0} = \psi_{m0}\alpha_{PM0}$, and

$$g_{\alpha}(\gamma', \beta'_0) = \frac{\bar{i}'_{q\alpha} \bar{\omega}_{\alpha}^2 \beta'_0}{(1 + \alpha_0(\bar{\Theta}_{\alpha} - 20))^{\gamma'}}.$$

This model has 5 unknown parameters to be identified (R_{DC0} , β'_0 , γ' , ψ_{m0} , α'_{PM0}) and is nonlinear on these parameters. This model can be simplified in the affine model shown below:

$$\mathbf{y} = \mathbf{X}\phi + \mathbf{b}, \quad (36)$$

with

$$\mathbf{y} = \begin{bmatrix} \bar{u}'_{q1} \\ \vdots \\ \bar{u}'_{q\alpha} \\ \vdots \\ \bar{u}'_{qN_{OC}} \end{bmatrix}, \quad \phi = \begin{bmatrix} R_{DC0} \\ \gamma' \\ \beta'_0 \\ \psi_{m0} \\ \alpha'_{PM0} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_\alpha \\ \vdots \\ b_{N_{OC}} \end{bmatrix},$$

$$\mathbf{X} = \begin{bmatrix} \bar{i}'_{q1} & p_1 & q_1 & \bar{\omega}_1 & \bar{\omega}_1(\bar{\Theta}_1 - 20) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \bar{i}'_{q\alpha} & p_\alpha & q_\alpha & \bar{\omega}_\alpha & \bar{\omega}_\alpha(\bar{\Theta}_\alpha - 20) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \bar{i}'_{qN_{OC}} & p_{N_{OC}} & q_{N_{OC}} & \bar{\omega}_{N_{OC}} & \bar{\omega}_{N_{OC}}(\bar{\Theta}_{N_{OC}} - 20) \end{bmatrix},$$

where p_α , q_α and b_α are coefficients obtained from the first-order Taylor approximation of the function $g_\alpha(\gamma', \beta'_0)$ on the point $(\bar{\gamma}', \bar{\beta}'_0)$, with $\bar{\gamma}' = 1.5$ and $\bar{\beta}'_0 = \beta'_0/2$, which is chosen according to the considerations reported in Subsection III-D.

The LS finds the solution $\hat{\phi}$ that minimizes the voltage prediction error on the model (36):

$$\hat{\phi} = (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{b}). \quad (37)$$

The parameters R'_{AC} and ψ_m in the OC α are obtained as follows:

$$\hat{R}'_{AC\alpha} = \frac{(\mathbf{X})_{\alpha,1:3}(\hat{\phi})_{1:3}}{\bar{i}'_{q\alpha}} + b_\alpha, \quad (38)$$

$$\hat{\psi}_{m\alpha} = \frac{(\mathbf{X})_{\alpha,4:5}(\hat{\phi})_{4:5}}{\bar{\omega}_\alpha},$$

where $(\mathbf{X})_{\alpha,i:j}$ expresses the elements from the i -th to the j -th rows of the α -th column of $\mathbf{X}$. To compare these two methods with the one proposed in this paper, an analysis considering scenarios with different number of available steady states is reported. The performances are compared over a set of 20 OCs, i.e. the first OCs of each steady state (see Fig. 5). As in the analysis shown in Fig. 10, for each OC, a number of trials provided by (32) are performed. Fig. 11 shows the MAPE and the maximum APE of the estimation of ψ_m when 2, 3 and 4 steady states are available. The black circles in the upper figure represent the estimations obtained with the FP method. Note that accuracy decreases with the increase of the motor speed as the variation of R with the frequency is not taken into account by this method. Moreover, also the voltage errors cannot be compensated with the FP method. Therefore, the proposed method outperforms the FP method almost in all the OCs even with few available steady states. At the top of the two subfigures are reported the estimations with MAPE and maximum APE higher than 3% and 8%, respectively. In these cases, a numeric value is added to specify the errors. Clearly, the LS fails when only 2 steady states are available since the number of the OCs is insufficient to correctly estimate $\hat{\phi}$.

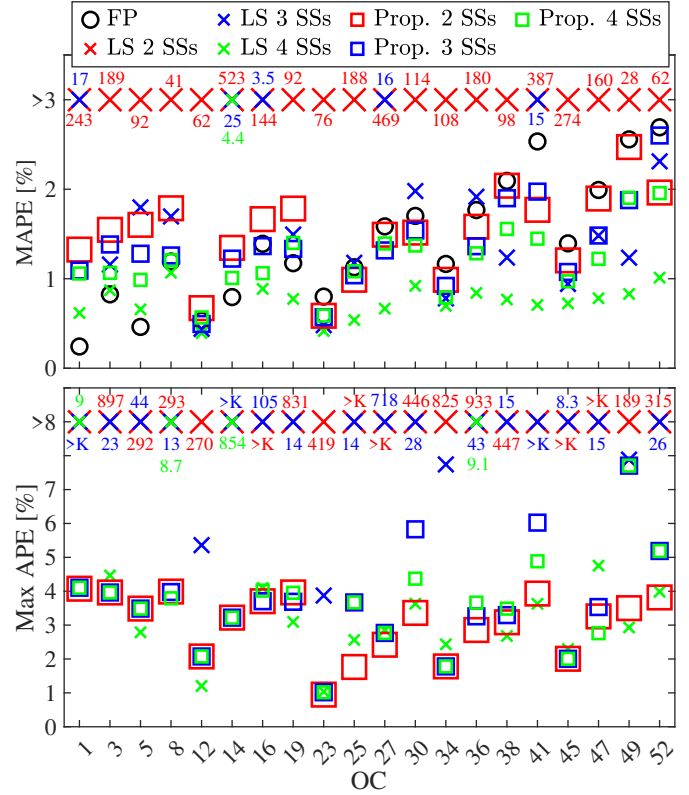


Fig. 11: Estimation errors of ψ_m with different methods and with 2, 3 and 4 available steady states. The numbers at the top of the subfigures denote percentage errors $> 3\%$ and $> 8\%$.

Instead, the proposed method provides reliable estimations with low maximum APE. When 3 steady states are available, the proposed method is still more accurate than the LS (for most of the OCs, the blue squares are below the blue crosses). Finally, when 4 steady states are available, the MAPEs obtained with the LS are lower than those of the proposed method. Instead, the maximum APEs are greater for the LS than for the proposed method in most of the OCs.

Fig. 12 reports the same analysis for R'_{AC} . Also in this case, the FP method is poorly accurate and is outperformed by the proposed method. This is mainly caused by the fact that this method does not take into account the variation of ψ_m with the temperature. In this figure, the symbol $> K$ that appears on the top of the subfigures denotes that the error is greater than 1000%. Moreover, the symbol $|R|$ denotes that the estimation has been rejected by the proposed method. As concerning the MAPE, with the same available steady states, the LS is on average outperformed by the proposed method in all the cases. Instead, as concerning the maximum APE, the gap between the LS and the propose method is even bigger. This is mainly due to the fact that the proposed method rejects the estimations affected by a high error majorant.

Fig. 13 shows the results of the estimations of ψ_m when 5, 7 and 9 steady states are available. The results obtained with the FP method are the same reported in Fig. 11. As concerning the MAPE, the LS outperforms the proposed method when 5 or 7 steady states are available, while the performances of the two methods are quite similar when 9 steady states are available.

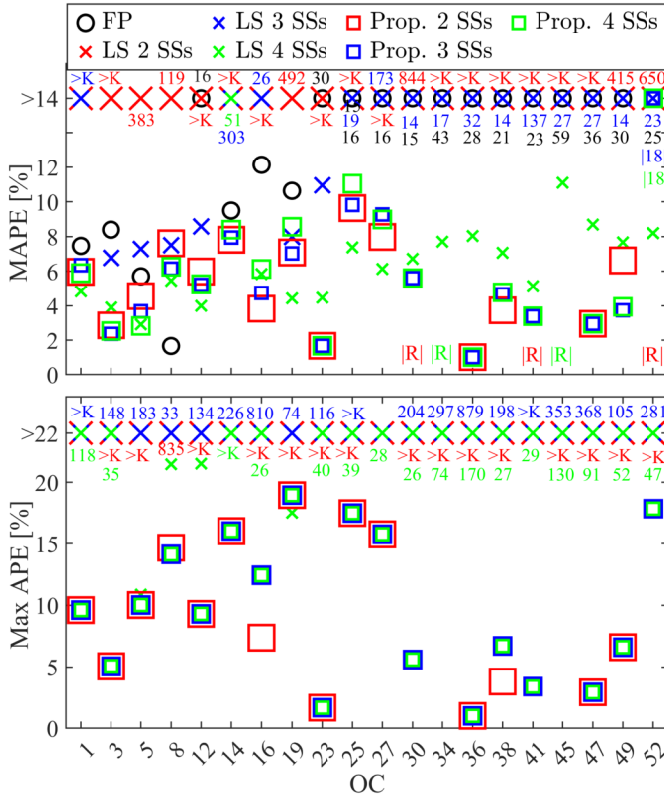


Fig. 12: Estimation errors of R'_{AC} with different methods and with 2, 3 and 4 available steady states. The numbers at the top of the subfigures denote percentage errors $> 14\%$ and $> 22\%$.

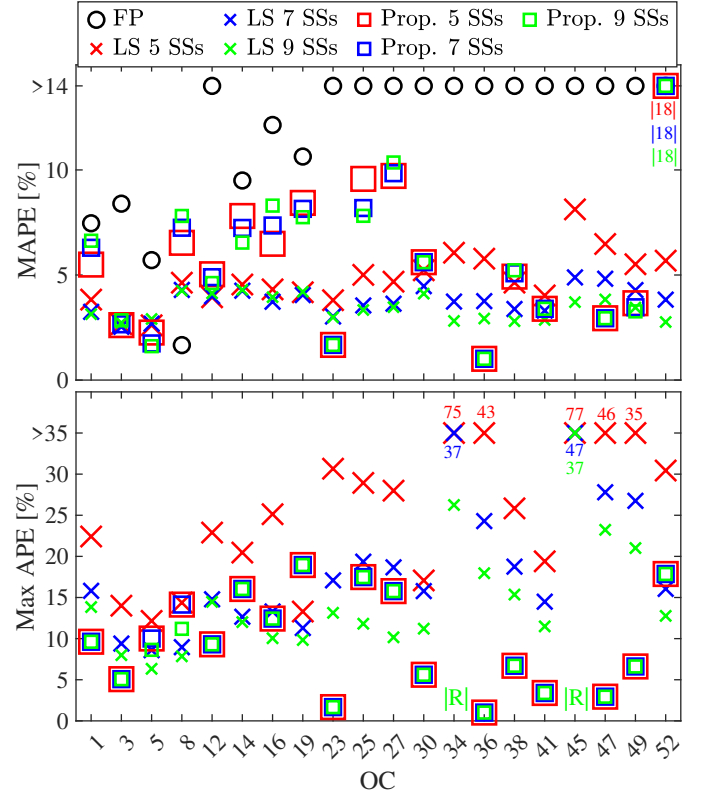


Fig. 14: Estimation errors of R'_{AC} with different methods and with 5, 7 and 9 available steady states. The numbers at the top of the subfigures denote percentage errors $> 14\%$ and $> 35\%$.

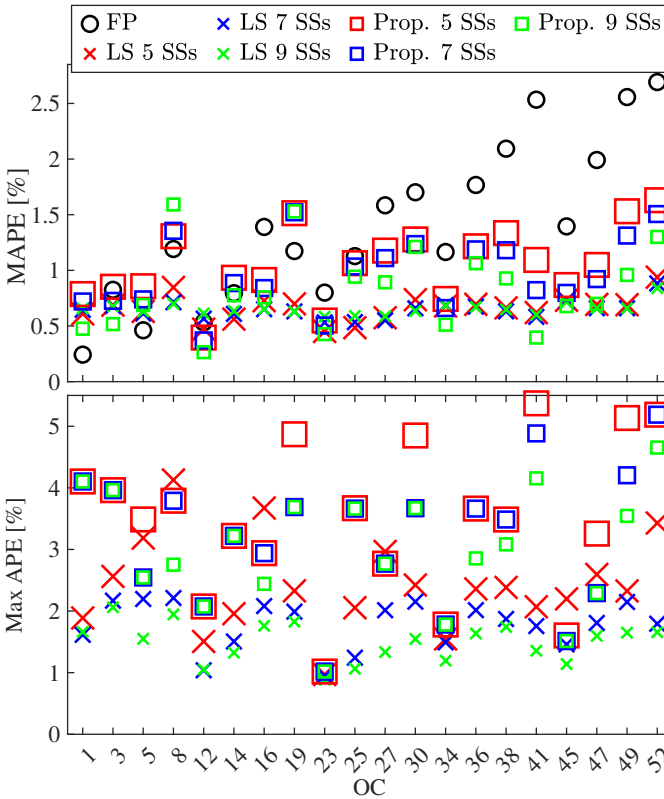


Fig. 13: Estimation errors of ψ_m with different methods and with 5, 7 and 9 available steady states.

Instead, in all the cases, the maximum APEs obtained with the proposed method are higher if compared with the LS.

Finally, Fig. 14 reports the same analysis for R'_{AC} . Note that although the LS generally achieves lower MAPEs across all three scenarios on average, the proposed method, on the other hand, exhibits significantly lower maximum APEs. It is worth noting that with the LS method, even when significant parameter estimation errors occur, the voltage prediction errors remain close to zero. Therefore, a criterion to reject estimations affected by high voltage prediction errors would be ineffective. Instead, the proposed method ensures to limit the maximum estimation error by taking into account the voltage errors and the parameter variations.

VI. EXPERIMENTAL RESULTS

This section reports the results obtained with the experimental dataset. Fig. 15 shows the results of the identification of L in all the 55 OCs using the d -axis compensated voltage and the d -axis voltage reference ($\hat{L}_\alpha^*$). The estimations are compared with the measured value. Note that the estimations obtained with the compensated voltage are very accurate and, as expected, the identified L does not significantly vary with both current and speed. Instead, the estimations obtained with the voltage reference worsen with the increase of the speed, as in the simulation analysis.

Regarding the identification of ψ_m and R'_{AC} , since only measurements of these parameters at room temperature are

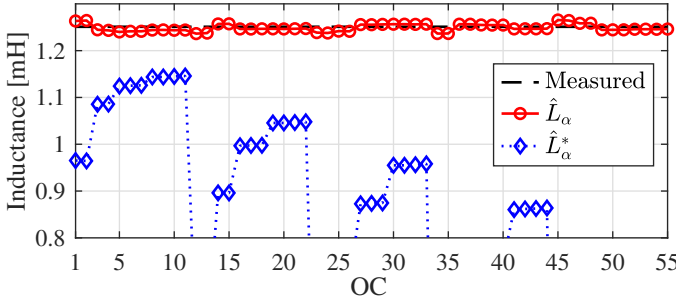


Fig. 15: Estimation of L .

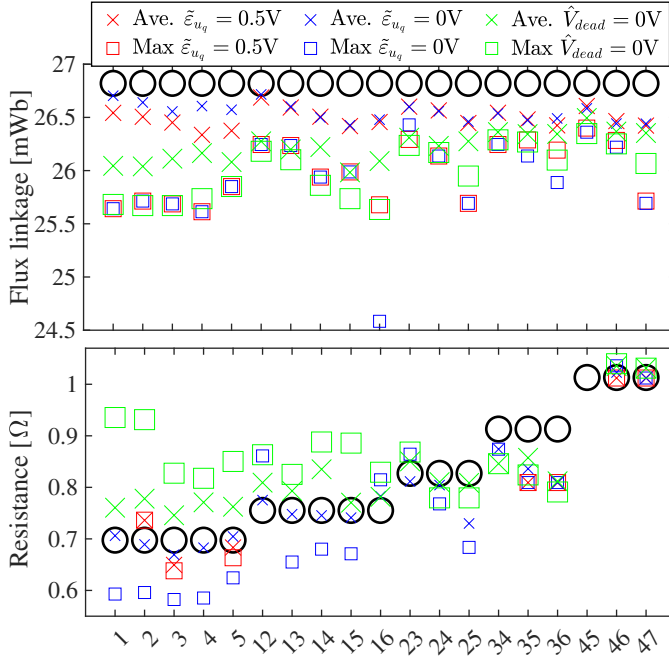


Fig. 16: Estimation of ψ_m and R'_{AC} with 5 steady states. The black circles represent measured values.

available, to properly evaluate the estimation accuracy, Fig. 16 shows the results obtained on a reduced subset of OCs, i.e. the OCs at temperature lower than 50 °C. In particular, the results have been obtained considering a scenario with 5 available steady states. The available steady states are randomly chosen among the full dataset and a number of trials equal to $N_5 = N_{comb_5}/5$ is evaluated. Since the maximum temperature difference with respect to the room temperature is 30 °C, the variation of the actual ψ_m and R'_{AC} can be neglected. The analysis is performed in three different cases, i.e. when setting $\tilde{\varepsilon}_{u_q} = 0.5V$, $\tilde{\varepsilon}_{u_q} = 0V$, and $\tilde{\varepsilon}_{u_q} = 0.5V$ with $\hat{V}_{dead} = 0V$. The third case allows to analyze the estimation accuracy when the VSI nonlinearity is not compensated. In particular, the figure shows the average and worst values of the estimations obtained with the N_5 trials. As concerns the estimation of ψ_m , the average estimations are very similar in the two first cases, with a high accuracy. However, in some OCs, the worst estimations obtained in the first case are more accurate than in the second one, confirming the importance of considering the voltage error to minimize the error majorant.

Note also that the average estimations slightly decrease with the increase of the OCs temperature (see Fig. 5). This aspect highlights that the proposed method can detect slightly rotor flux linkage variations with the temperature. Also the average estimations of R'_{AC} are highly accurate and can precisely track the resistance variations with the speed of the motor. In the first case ($\tilde{\varepsilon}_{u_q} = 0.5V$), the estimations are accepted only in few OCs, in a similar way to the simulation results shown in Fig. 8. Furthermore, the considerations drawn on the comparison between the average and worst estimations of ψ_m in the two first cases can also be repeated about the estimation of R'_{AC} . Finally, this analysis demonstrates the importance of estimating the distorted voltage. In fact, in most of the OCs the estimation accuracy decreases for both ψ_m and R'_{AC} when the VSI nonlinearity is not compensated.

VII. CONCLUSION

This paper has investigated the possibility of estimating the parameters of an isotropic permanent magnet synchronous machine only using the measurements commonly available during the regular operation of commercial drives. The rank-deficiency issue, typical of this class of problem, is solved by using two different operating conditions. In particular, by properly selecting these two operating conditions from the available dataset of measurements it is possible to minimize the estimation errors due to both parameter variations and voltage errors. The effectiveness of the proposed identification approach has been validated using both simulation and experimental studies considering several scenarios with different number of available speed/load operating conditions. The key findings of the manuscript are summarized as follows: 1) the accuracy of the estimations of both ψ_m and R improves when the q -axis voltage error component is considered when selecting the optimal operating conditions required to perform their identification; 2) the accuracy of the estimations improves when more steady states are available; 3) the number of identifiable operating condition increases with the size of the available dataset of measurements. Also, the proposed method has been compared with other methods from the literature. The results clearly show that the proposed method outperforms the method based on fixing a parameter to its nominal value. Additionally, the proposed approach works better than the least squares method in scenarios with a limited number of steady states. Although, on average, the least squares technique performs better than the proposed one in scenarios with more than four available steady-states, the estimations of the stator resistance produced by the least square method are impacted by larger maximum errors. Hence, the proposed method is more flexible as provides accurate estimations even in scenarios with few available steady states and is more cautionary as it autonomously reject estimations affected by high errors. These outcomes demonstrate the superiority of the proposed identification method to any application where additional sensor, dedicated tests and signal injections are not possible. Such feature makes the proposed method highly suitable for the condition monitoring of large-scale industrial plants where many electrical machines are operating at the same time.

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